

APARNA KRISHNAN

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Education

Virginia Polytechnic Institute and State University, Blacksburg, VA

August 2023 – Present

PhD. Theoretical Chemistry

Advisor : Prof T Daniel Crawford

3.82/4.0

Honors: Phi Kappa Phi

Experience: Time-dependent coupled cluster, Linear response theory, Spectroscopy, Diagonalization algorithms, Analytic gradient theory, Numerical differentiation, Scientific automation pipelines

Birla Institute of Technology and Science Pilani, Goa

August 2018 – May 2023

Int. MSc.(Hons.) Chemistry and B.E.(Hons.) Mechanical Engineering

9.01/10.0

Honors: Rank 1, Chemistry

Coursework: Optimization, Group theory, Statistical Mechanics, Quantum Mechanics, Quantum Chemistry and Spectroscopy, Electronic Structure Theory, Finite Element Methods, Numerical Methods for Software Mathematics, Machine Learning

Publications

[†]These authors contributed equally.

Chemical Interpretation of Time-Dependent Coupled-Cluster Theory

A. Krishnan, H. E. Kristiansen, B. G. Peyton, T. D. Crawford, T. B. Pedersen

Journal of Physical Chemistry A - **accepted**. *arXiv:2605.17409*

Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism within Configuration Interaction Theory

A. Krishnan, B. Shumberger, T. D. Crawford

Journal of Chemical Theory and Computation - *submitting 2026*

Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism at the Coupled-Cluster Singles and Doubles Level

A. Krishnan, T. D. Crawford

In preparation

Investigating Correlation in Analytic Vibrational Circular Dichroism: HF, MP2, CISD, and CCSD

A. Krishnan, T. D. Crawford

In preparation

Real-Time Unitary Coupled-Cluster Methods for Molecular Spectroscopy

A. Melekamburath[†], **A. Krishnan**[†], E. F. Valeev, T. D. Crawford

In preparation

PyCC: An Open-Source Python-Based Coupled Cluster Package

AUTHOR1, **A. Krishnan**, AUTHORS, T. D. Crawford

In preparation

Deformation Characterization of Additive Manufactured Polyether Ether Ketone Using Mirror Assisted Image Correlation

V. Gupta, **A. Krishnan**, N. I. Thiruselvam, D. M. Kulkarni, V. V. Chaudhari, S. Suraj, R. Das

Submitted

Research Experience

Virginia Polytechnic Institute and State University

August 2023 – Present

Graduate Research Assistant, Advisor: Prof T Daniel Crawford

Blacksburg, USA

- Developed analytic vibrational circular dichroism theory at the CISD, MP2, and CCSD levels via a density-matrix reformulation of atomic axial tensors; proved a 2-RDM vanishing theorem for the orbital-orbital AAT block that removes an entire class of terms.
- Implemented complete analytic pipelines (Hessians, atomic polar tensors, atomic axial tensors, rotational strengths), validated against finite difference to better than 10^{-7} .

- Showed that real-time and equation-of-motion unitary coupled-cluster excitation energies are not equivalent at finite BCH truncation; derived the discrepancy in closed form and certified it against exact operator algebra in an exactly solvable model.
- Contributor to PyCC (16 merged commits): implemented TD-EOM-CC methods, analytic CISD-VCD APTs/AATs/Hessians, CISD polarizabilities via $2n+1$ and explicit routes, and orbital decomposition analysis.

Virginia Polytechnic Institute and State University

August 2022 – May 2023

Thesis Intern, Advisor: Prof T Daniel Crawford

Blacksburg, USA

- Automated the spin integration of coupled cluster equations and generation of code, reducing manual calculation time by 80 percent and allowing research team to focus on complex quantum chemistry challenges.
- Formulated a multi-scale approach, combining MD and DFT, and devised a Python script that automated MD frame sampling; enhanced Raman spectra generation, and solvent shell definition, leading to a significant reduction in runtime from 10 days to 12 hours.

Indian Institute of Technology, Bombay

May 2021 – March 2022

Research Intern, Advisor: Prof Rahul Maitra

Mumbai, India

- Established a rigorous theoretical understanding of CC theory and explored the applications in quantum chemistry.
- Developed an innovative engine that automates Wick's theorem. Also implemented a fully automatic equation and code generator for dual exponent formulation of CC theory involving scattering-type operators with minimal user input.

Dhio Research and Engineering Pvt Ltd.

May - July 2020

Research Intern (Industry), Advisor: Santosh NL

Bangalore, India

- Simulated the visco-elastic properties of polymeric materials using J-OCTA and proposed processing and performance improvements for rubber-type polymers in the automotive industry; identified critical polymer architectures for better on-road tire performance, with ab-initio calculations 5x faster than the existing baseline.

Software

vcd-kit: analytic vibrational circular dichroism at the HF, MP2, CISD, and CCSD levels - atomic axial tensors, polar tensors, Hessians, and rotational strengths from a density-matrix formulation.

PyCC: open-source Python coupled-cluster package. Contributor (16 merged commits) - TD-EOM-CC, VCD pipelines, orbital decomposition analysis.

Biorthogonal Jacobi-Davidson: non-Hermitian eigensolver tracking left and right eigenvectors with enforced biorthogonality.

CC Equation Generator: symbolic derivation of dual-exponent CC equations with automatic `np.einsum` code generation.

Solvated Spectra Pipeline: automated MD-to-DFT workflow for chiroptical spectra of solvated molecules; 10 days to 12 hours.

Projects

Density-based reformulation of correlated vibrational circular dichroism

- Originated a density-matrix reformulation of atomic axial tensors (AATs) after recognising that machinery developed for the diagonal Born-Oppenheimer correction could be transferred to chiroptical response, and built it into a single framework spanning HF, MP2, CISD, and CCSD.
- Proved a 2-RDM vanishing theorem for the orbital-orbital AAT block, removing an entire class of terms and making the framework cleanly extensible to coupled cluster.
- Implemented the complete analytic pipeline at each level - Hessians, normal-mode analysis, atomic polar tensors, atomic axial tensors, rotational strengths - validated against finite difference to better than 10^{-7} . Used the Z-vector/Lagrangian approach at MP2 to avoid solving coupled-perturbed equations per nuclear displacement.
- Extended the framework with gauge-including atomic orbitals (GIAOs) for origin-independent AATs, improving accuracy for basis-set-dependent magnetic response properties.

Real-time coupled cluster methods for the calculation of field-dependent molecular properties

- Revealed the orbital and symmetry contribution to peaks in an absorption spectra. Crafted an algorithm using autocorrelation as a sole metric for deriving field-dependent properties and orbital decomposition, which can reduce storage requirements by 95 percent.

Structural non-equivalence of real-time and EOM unitary coupled cluster

- Showed that the RT/EOM equivalence guaranteed in conventional coupled cluster fails for the unitary ansatz at finite BCH truncation; derived the discrepancy in closed form, proved it vanishes in the two limits recovering the standard result, and certified it against brute-force operator algebra to 10^{-12} .

A bi-orthogonal Jacobi-Davidson method to solve the eigenvectors of a non-Hermitian system

- Engineered a bi-orthogonal Jacobi-Davidson (biJD) framework that maintains bi-orthogonality between left and right eigenvectors throughout, converging roughly 25x faster than a conventional Jacobi-Davidson implementation on the same test problem.

Mirror-assisted Digital Image Correlation (DIC) to extract side surface information

- Designed and validated a mirror-assisted 3D-DIC technique for measuring side surface deformations in 3D printed carbon-reinforced PEEK, enhancing accuracy and reducing equipment complexity.

Skills

Languages: Python, C++

Scientific Computing: NumPy, SciPy, SymPy (secondquant), einsum, h5py/HDF5, Pandas, Matplotlib, Jupyter

Quantum Chemistry: Psi4, Gaussian 16, CFOUR, PyCC

HPC & Tooling: SLURM, ARC cluster (Virginia Tech), Linux/Bash, Git/GitHub, L^AT_EX

Quantum Computing: Qiskit, Cirq, QDK, quantum algorithms

Machine Learning: PyTorch, TensorFlow, Keras, NLP

Certifications: Quantum Computing (QWorld), Machine Learning (Stanford), Quantum Excellence (QGSS'25), Quantum Error Correction (Google QAI, in progress)

Talks and Presentations

MES 2026, Greece - Density-Matrix Formulation of Atomic Axial Tensors for Vibrational Circular Dichroism within Correlated theories. *Upcoming, September 2026.*

SETCA 2026, Georgia State University, Atlanta - Density-matrix formulation of atomic axial tensors for CISD vibrational circular dichroism. **Best Poster Award.**

MRPSS 2025, Stockholm, Sweden - A tale of two decompositions: analyzing electronic excitations in real-time coupled-cluster.

SETCA 2025, University of Memphis, Tennessee - Orbital-resolved absorption spectra from RT-CC autocorrelation function.

SERMACS 2024, Atlanta, Georgia - Understanding real-time coupled-cluster absorption spectra: orbital contributions from individual transitions.

ESQC 2024, Palermo, Italy - Understanding real-time coupled-cluster absorption spectra: orbital contributions from individual transitions.

SETCA 2023, University of South Carolina, Columbia - Determination of optical properties by modeling complex solvent effects: molecular dynamics and density functional approach.

Achievements

Best Poster Award, SETCA 2026: Awarded for the density-matrix AAT formulation of CISD vibrational circular dichroism, Atlanta.

KVPY Scholar, 2019: Highly competitive national scholarship awarded by the Department of Science and Technology, Government of India, to students pursuing a career in science.

Elected member of Phi Kappa Phi: Elected for outstanding academic standing at Virginia Tech.

BITS Pilani Travel Award: Awarded to complete the Bachelor's thesis at a foreign university.